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FIAP — Pós Tech **Software Architecture**

Evidências de Escalabilidade

Kubernetes e execução em ambientes

Minikube, HPA, CI/CD, AWS e integração de e-mail

Oficina Mecânica — Fase 2

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1. Kubernetes e escalabilidade

O mesmo sistema foi implantado no Minikube para validar sua execução em Kubernetes. O cluster utilizou Kubernetes 1.31, metrics-server, PostgreSQL, aplicação, Service e HPA.

```
Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube start `
>> --driver=docker `
>> --cpus=2 `
>> --memory=4096 `
>> --kubernetes-version=v1.31.0
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.31.0 preload ...
|
```

Criação do cluster Minikube.

```
Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 29s v1.31.0
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> |
```

Cluster ativo e node em estado Ready.

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1.1 Implantação da aplicação

A imagem Docker foi construída para o ambiente do Minikube. Depois foram aplicados os manifestos de configuração, banco, aplicação, serviço e autoscaling.

```
Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube addons enable metrics-server
* metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
- Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
* The 'metrics-server' addon is enabled
PS C:\fiap-tech-challenge-oficina-mecanica-fase2>
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- rollout status deployment/metrics-server `
>> -n kube-system `
>> --timeout=180s
Waiting for deployment "metrics-server" rollout to finish: 0 of 1 updated replicas are available...
deployment "metrics-server" successfully rolled out
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> |
```

metrics-server habilitado para fornecer as métricas do cluster.

```
Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- apply -f .\k8s\app-service.yaml
service/oficina-app created
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- apply -f .\k8s\app-deployment.yaml
deployment.apps/oficina-app created
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- rollout status deployment/oficina-app `
>> -n oficina `
>> --timeout=300s
Waiting for deployment "oficina-app" rollout to finish: 0 of 2 updated replicas are available...
Waiting for deployment "oficina-app" rollout to finish: 1 of 2 updated replicas are available...
deployment "oficina-app" successfully rolled out
PS C:\fiap-tech-challenge-oficina-mecanica-fase2>
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- apply -f .\k8s\hpa.yaml
horizontalpodautoscaler.autoscaling/oficina-app-hpa created
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> |
```

Aplicação, Service e HPA aplicados com rollout concluído.

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```

Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- get pods,svc,hpa,pvc -n oficina
NAME                                READY  STATUS   RESTARTS  AGE
pod/oficina-app-777dbb895c-mmdtn    1/1    Running  0          87s
pod/oficina-app-777dbb895c-xg4fx    1/1    Running  0          87s
pod/oficina-db-569469fff-js7mq      1/1    Running  0          2m28s

NAME                                TYPE               CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
service/oficina-app                 LoadBalancer      10.106.171.141 <pending>    80:32705/TCP    87s
service/oficina-db                  ClusterIP           10.104.169.255 <none>       5432/TCP        2m30s

NAME                                REFERENCE          TARGETS          MINPODS  MAXPOD
horizontalpodautoscaler.autoscaling/oficina-app-hpa  Deployment/oficina-app  cpu: <unknown>/70%, memory: 42%/80%  2        5

NAME                                STATUS  VOLUME          CAPACITY  ACCESS MODES  STORAGECLASS  V
OLUETATTRIBUTESCLASS  AGE
persistentvolumeclaim/oficina-db-pvc  Bound   pvc-1b58a4ab-e9c5-4645-9487-6141df5e33e7  5Gi        RWO           standard      <
unset>                                2m30s
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> |
  
```

Pods, Services, HPA e volume persistente no namespace oficina.

```

Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> Test-NetConnection localhost -Port 8080

ComputerName : localhost
RemoteAddress : ::1
RemotePort : 8080
InterfaceAlias : Loopback Pseudo-Interface 1
SourceAddress : ::1
TcpTestSucceeded : True

PS C:\fiap-tech-challenge-oficina-mecanica-fase2> Invoke-RestMethod http://localhost:8080/actuator/health

status groups
-----
UP      {liveness, readiness}

PS C:\fiap-tech-challenge-oficina-mecanica-fase2> Start-Process http://localhost:8080/swagger-ui/index.html
  
```

Aplicação no Minikube respondendo ao health check.

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1.2 Teste do HPA

O HPA foi configurado com duas réplicas mínimas, cinco máximas, alvo de 70% de CPU e 80% de memória. Antes da carga, as métricas estavam disponíveis e o HPA estava ativo.

```
Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> Start-Sleep -Seconds 45
PS C:\fiap-tech-challenge-oficina-mecanica-fase2>
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- top nodes
NAME          CPU(cores)  CPU%    MEMORY(bytes)  MEMORY%
minikube      176m       1%      1949Mi         12%
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- top pods -n oficina
NAME          CPU(cores)  MEMORY(bytes)
oficina-app-777dbb895c-mmdtn  4m          296Mi
oficina-app-777dbb895c-xg4fx  4m          274Mi
oficina-db-569469fff-js7mq    3m          107Mi
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- get hpa oficina-app-hpa -n oficina
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
oficina-app-hpa  Deployment/oficina-app  cpu: 1%/70%, memory: 55%/80%  2        5        2        10m
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- describe hpa oficina-app-hpa -n oficina
Name:
Namespace:
Labels:
Annotations:
CreationTimestamp:
Reference:
Metrics:
  resource cpu on pods (as a percentage of request): 1% (4m) / 70%
  resource memory on pods (as a percentage of request): 55% (298962944) / 80%
Min replicas: 2
Max replicas: 5
Behavior:
  Scale Up:
    Stabilization Window: 60 seconds
    Select Policy: Max
    Policies:
      - Type: Pods Value: 1 Period: 60 seconds
  Scale Down:
    Stabilization Window: 300 seconds
    Select Policy: Max
    Policies:
      - Type: Pods Value: 1 Period: 120 seconds
```

Linha de base do HPA com duas réplicas e métricas de CPU e memória.

```
Windows PowerShell
Name: oficina-app-hpa
Namespace: oficina
Labels: app.kubernetes.io/name=oficina-app, app.kubernetes.io/part-of=oficina-backend
Annotations:
CreationTimestamp: Sun, 12 Jul 2026 11:38:35 -0300
Reference: Deployment/oficina-app
Metrics:
  resource cpu on pods (as a percentage of request): 1% (4m) / 70%
  resource memory on pods (as a percentage of request): 55% (298962944) / 80%
Min replicas: 2
Max replicas: 5
Behavior:
  Scale Up:
    Stabilization Window: 60 seconds
    Select Policy: Max
    Policies:
      - Type: Pods Value: 1 Period: 60 seconds
  Scale Down:
    Stabilization Window: 300 seconds
    Select Policy: Max
    Policies:
      - Type: Pods Value: 1 Period: 120 seconds
Deployment pods: 2 current / 2 desired
Conditions:
  Type          Status Reason Message
  ---
AbleToScale    True  ReadyForNewScale recommended size matches current size
ScalingActive  True  ValidMetricFound the HPA was able to successfully calculate a replica count from memory resource utilization (percentage of request)
ScalingLimited False DesiredWithinRange the desired count is within the acceptable range
Events:
  Type          Reason          Age          From          Message
  ---
Warning FailedGetResourceMetric 9m33s (x6 over 10m) horizontal-pod-autoscaler failed to get cpu utilization: did not receive metrics for targeted pods (pods might be unready)
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> |
```

Condições AbleToScale e ScalingActive confirmadas.

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Foram criados pods geradores de carga. A CPU ultrapassou o alvo e o HPA aumentou progressivamente as réplicas da aplicação, chegando a quatro réplicas nas capturas registradas, dentro do limite máximo configurado de cinco.

```
Windows PowerShell
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> 1..10 | ForEach-Object {
>> minikube kubectl -- run "load-generator-$_" `
>> -n oficina `
>> --image=busybox:1.36 `
>> --restart=Never `
>> -- /bin/sh -c "while true; do wget -q -O- http://oficina-app/actuator/health > /dev/null; done"
>> }
pod/load-generator-1 created
pod/load-generator-2 created
pod/load-generator-3 created
pod/load-generator-4 created
pod/load-generator-5 created
pod/load-generator-6 created
pod/load-generator-7 created
pod/load-generator-8 created
pod/load-generator-9 created
pod/load-generator-10 created
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> |
```

Criação dos pods geradores de carga.

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
oficina-app-hpa	Deployment/oficina-app	cpu: 1%/70%, memory: 55%/80%	2	5	2	13m
oficina-app-hpa	Deployment/oficina-app	cpu: 1%/70%, memory: 55%/80%	2	5	2	13m
oficina-app-hpa	Deployment/oficina-app	cpu: 1%/70%, memory: 56%/80%	2	5	2	14m
oficina-app-hpa	Deployment/oficina-app	cpu: 1%/70%, memory: 56%/80%	2	5	2	15m
oficina-app-hpa	Deployment/oficina-app	cpu: 67%/70%, memory: 59%/80%	2	5	2	16m
oficina-app-hpa	Deployment/oficina-app	cpu: 185%/70%, memory: 59%/80%	2	5	2	17m
oficina-app-hpa	Deployment/oficina-app	cpu: 185%/70%, memory: 59%/80%	2	5	2	18m
oficina-app-hpa	Deployment/oficina-app	cpu: 134%/70%, memory: 59%/80%	2	5	3	18m
oficina-app-hpa	Deployment/oficina-app	cpu: 134%/70%, memory: 59%/80%	2	5	3	19m
oficina-app-hpa	Deployment/oficina-app	cpu: 74%/70%, memory: 51%/80%	2	5	4	19m

Aumento progressivo de duas para três e quatro réplicas durante a carga.

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```
Windows PowerShell x Windows PowerShell + v
NAME CPU(cores) MEMORY(bytes)
load-generator-1 53m 0Mi
load-generator-10 52m 1Mi
load-generator-2 51m 0Mi
load-generator-3 51m 0Mi
load-generator-4 53m 0Mi
load-generator-5 53m 0Mi
load-generator-6 53m 0Mi
load-generator-7 51m 0Mi
load-generator-8 52m 0Mi
load-generator-9 52m 0Mi
oficina-app-777dbb895c-hklhd 415m 280Mi
oficina-app-777dbb895c-mmdtn 121m 314Mi
oficina-app-777dbb895c-xg4fx 126m 296Mi
oficina-app-777dbb895c-zn5nk 412m 185Mi
oficina-db-569469fff-js7mq 33m 121Mi
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- get hpa -n oficina
NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
oficina-app-hpa Deployment/oficina-app cpu: 49%/70%, memory: 52%/80% 2 5 4 21m
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- get pods -n oficina
NAME READY STATUS RESTARTS AGE
load-generator-1 1/1 Running 0 5m25s
load-generator-10 1/1 Running 0 5m23s
load-generator-2 1/1 Running 0 5m25s
load-generator-3 1/1 Running 0 5m24s
load-generator-4 1/1 Running 0 5m24s
load-generator-5 1/1 Running 0 5m24s
load-generator-6 1/1 Running 0 5m24s
load-generator-7 1/1 Running 0 5m24s
load-generator-8 1/1 Running 0 5m24s
load-generator-9 1/1 Running 0 5m23s
oficina-app-777dbb895c-hklhd 1/1 Running 0 3m
oficina-app-777dbb895c-mmdtn 1/1 Running 0 22m
oficina-app-777dbb895c-xg4fx 1/1 Running 0 22m
oficina-app-777dbb895c-zn5nk 1/1 Running 0 119s
oficina-db-569469fff-js7mq 1/1 Running 0 23m
PS C:\fiap-tech-challenge-oficina-mecanica-fase2>
```

Quatro réplicas em execução com consumo de CPU distribuído.

```
Windows PowerShell x Windows PowerShell + v
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> 1..10 | ForEach-Object {
>> minikube kubectl -- delete pod "load-generator-$_" `
>> -n oficina `
>> --ignore-not-found
>> }
pod "load-generator-1" deleted
pod "load-generator-2" deleted
pod "load-generator-3" deleted
pod "load-generator-4" deleted
pod "load-generator-5" deleted
pod "load-generator-6" deleted
pod "load-generator-7" deleted
pod "load-generator-8" deleted
pod "load-generator-9" deleted
pod "load-generator-10" deleted
PS C:\fiap-tech-challenge-oficina-mecanica-fase2>
```

Remoção dos pods geradores de carga.

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Após o encerramento da carga, a aplicação permaneceu saudável. O HPA estabilizou temporariamente em três réplicas por causa da métrica de memória e da regra de arredondamento. Ao final, o namespace e o cluster foram removidos.

```

Windows PowerShell
=== HPA ===
NAME                REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
oficina-app-hpa     Deployment/oficina-app  cpu: 0%/70%, memory: 59%/80%    2         5         2        90m

=== PODS ===
NAME                READY  STATUS   RESTARTS  AGE
oficina-app-777dbb895c-mmdtn  1/1    Running   0         91m
oficina-app-777dbb895c-xg4fx  1/1    Running   0         91m
oficina-db-569469fff-js7mq    1/1    Running   0         92m
  
```

Estado saudável após a carga, com três réplicas estabilizadas.

```

PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube kubectl -- delete namespace oficina
namespace "oficina" deleted
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube stop
Stopping node "minikube" ...
Powering off "minikube" via SSH ...
1 node stopped.
PS C:\fiap-tech-challenge-oficina-mecanica-fase2> minikube delete -p minikube
Deleting "minikube" in docker ...
Deleting container "minikube" ...
Removing C:\Users\tiago\.minikube\machines\minikube ...
Removed all traces of the "minikube" cluster.
PS C:\fiap-tech-challenge-oficina-mecanica-fase2>
  
```

Remoção do namespace e encerramento do Minikube.